

Path Safety Assessment Tool (PSAT)

User Guide

Welcome to the Path Safety Assessment Tool (PSAT). This tool allows you to evaluate active mobility paths using the CycleRAP model.

1. Getting Started

- **Start a Project:** From the Home page, click “Create Project” to import your street-level imagery.
- **Project Settings:** Ensure your project name (e.g. Road Name) is correctly set. You can insert relevant tags (e.g. Ang Mo Kio) to group your projects. Upload your images by selecting the correct local drive.
- **Select project(s):** You can select single or manage multiple projects from the Projects listing. You can also edit the metadata (i.e. Project Name and Tags) of the project(s).

2. Coding Page

Auto-coding: Use the “Auto-code image” button to leverage AI models that automatically identify risk factors (e.g. facility type, obstacles on paths) from the images. There is an option to auto-code by image or all images within the project. The “Segments Autocoded” will automatically be updated.

- **Auto-coding algorithms:** Computer Vision and GIS Coding/Mapping:
- CV utilizes image learning algorithms to systematically identify patterns and features.
- Our GIS backend automatically evaluates contextual geo-referenced spatial data such as proximity of the images to MRT exits, bus stops, and road intersections.
- **Manual Review:** You can review and override the attributes predicted by the AI/GIS directly on the panel.
- You can review the attributes by different categories, then click “Next”, or jump between sections.
- You can refer to the Coding Guide for coding the various attributes.
- Under the “Details” tab, users can find the exact parameters of some attributes (e.g. path width, radius, and grade), and a zoomed-in map view of a 5m radius buffer overlaying the data source layers.
- A red box “Manual Edit Done” will appear to label attributes that have been manually overwritten. There is also an Autocode Validation table at the bottom of the page that summarizes the percentage change of overwritten attributes.
- After finishing the manual review, remember to update the “Segments Verified”.

3. Map View & Analysis

The Map View & Analysis section allows users to visualize, review, and edit path segments within the project using interactive map tools and GIS data layers.

- **GIS Layers:** You can toggle different map layers to visualize Footpaths, Cycling Paths, Road, crossings, bus stops, MRT stations, etc. This helps provide spatial context when reviewing segments.
- **Risk Bands:** Segments are color-coded based on overall risk (i.e. worst risk of any SB, BB, BP, VP category). This allows users to quickly identify higher-risk and lower-risk segments on the map.
- **Segment Selection:** You can click on individual segment on the map to view its image, associated attributes and risk levels. The selected segment will be highlighted for easier review.
- **Map Navigation:** You can zoom in and out and pan across the map to focus on specific areas or segments.
- **Editing:** You can select or delete segment points directly on the map preview using the cursor tools. This allows you to refine segment boundaries and improve accuracy.

4. Path Analysis

The Path Analysis module allows users to analyze projects and evaluate the safety risk levels of path segments.

- **Project Selection:** You can select one or more projects to analyze.
- **Filter Projects:** You can filter projects based on:
 - Project Name
 - Date Created
 - Last Updated
 - Tags
- **Filter Segment:** You can apply up to 5 filters to visualize segments with specific attributes. This allows you to focus on segments that meet certain conditions.
- **Risk Assessment:** The system evaluates segment attributes (e.g. line of sight, path width, surface condition) to determine risk levels for different crash types:
 - Bicycle–Bicycle
 - Bicycle–Pedestrian
 - Single Bicycle
 - Vehicle–Bicycle
- **Segment Visualization:** Filtered segments are displayed based on the selected criteria, allowing users to identify patterns and higher-risk areas.
- **Validation:** User can toggle back to the coding sheet and update the parameters.
- **Data export:** User can export the filtered data into CSV. file.

5. Treatment Application

The Treatment Application acts as a scenario-testing tool to see how safety interventions can reduce the baseline risk. It applies predefined risk multipliers and condition penalties for specific treatments to the chosen segments, recalculating the safety score to show potential improvements.

- **Apply Standard Treatments:** Select from a library of interventions, such as 'Widen Path', 'Improve Lighting', or 'Provide Physical Separation'.
- **Compare Before and After:** View side-by-side risk score reductions (e.g., seeing a Bicycle-Pedestrian crash type score drop after applying a segregated path treatment).
- **Batch Treatment Application:** Apply a selected treatment across multiple segments simultaneously to evaluate corridor-wide safety improvements.
- **Export Interventions:** Generate a report of the applied treatments and the estimated safety benefits to present to stakeholders and decision-makers.